Using Association Rule Learning to Identify Inventory Gaps in Wholesale Distribution Networks

by

Avi Walia

Supervisor: Professor Yuri Lawryshyn

Department of Computer Science
University of Toronto

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Avi Walia
Honours Bachelor of Science

Department of Computer Science
University of Toronto
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Abstract

This thesis presents a data-driven approach to improving inventory decision-making by identifying demand-driving primary products stocked without their related secondary products through the analysis of product co-purchasing patterns. This study develops a framework that uses historical transaction data, association rule learning (ARL), and clustering to identify relationships between products. Market basket analysis techniques, including the Apriori and Frequent Pattern Growth (FP-Growth) algorithms, are applied to extract recurring co-purchase patterns from transaction data. These patterns, together with clustering on product-level information, are used to construct mappings between primary products that generate independent demand and their associated secondary products that serve as essential complements. The framework is then combined with inventory and stocking data to identify situations where complementary products are not stocked together to flag items that may be at risk of obsolescence due to incomplete product pairings. The proposed approach demonstrates how the combination of ARL and clustering can be used to uncover hidden demand structures within historical transaction data and generate actionable insights to support more effective inventory management and supply chain optimization.

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Chapter 1

Introduction

1.1 The Problem and Opportunity

Managing inventory in large-scale distribution environments presents a complex challenge, particularly when products are treated as independent items rather than as part of a broader system of interrelated demand. In many cases, products are frequently purchased together, forming implicit relationships that are not captured in traditional inventory classification systems. The absence of this relational understanding can lead to inefficient stocking decisions, where complementary items are not available alongside one another. As a result, demand for certain products may be suppressed, leading to missed sales opportunities and an increased risk of premature obsolescence. Identifying and leveraging these hidden relationships, therefore, supports improved inventory performance and helps ensure that stocking strategies better reflect actual purchasing behaviour.

Prior research in inventory optimization shows that transaction-level product associations reveal latent demand structures that traditional inventory classification methods fail to capture (Hübner, 2011). One approach to addressing this challenge is to analyze historical transaction data to uncover patterns of co-purchasing. Association rule learning (ARL), a class of data mining techniques commonly used in market basket analysis, offers a systematic method for identifying these relationships. By applying algorithms such as Apriori and Frequent Pattern Growth (FP-Growth), it is possible to extract frequent item combinations and infer meaningful associations between products. These relationships can then be used to classify products into two functional categories: demand-generating (primary) or demand-dependent (secondary). This classification allows for a clearer understanding of product roles within the inventory, providing a foundation for more informed stocking decisions.

While market basket analysis is widely used to identify co-purchase relationships, its results are typically applied to promotion planning or recommendation systems rather than inventory classification. This research extends ARL into the inventory decision-making domain by transforming transactional dependencies into product-role mappings that support stocking strategies and obsolescence risk identification.

The industry partner that supplied the proprietary dataset for this study is a wholesale distributor specializing in HVACR, plumbing, and industrial equipment. Managing a large and diverse inventory across North America, the company faces challenges related to lost revenue and inventory inefficiencies resulting from suboptimal stocking decisions.

1.2 Objective

The primary objective of this thesis is to develop a data-driven framework that utilizes historical transaction data and association rule learning to classify products into a one-to-many mapping of primary and secondary products, ultimately improving stocking decisions and reducing the risk of product obsolescence.

1.3 Contributions

The following contributions were delivered to the company.

- **Python Code:** Implementation of association rule learning algorithms (Apriori or FP-Growth), along with an analytical framework to compare learned product relationships with inventory data and identify items currently at risk of obsolescence.
- **Sparks Table:** Cleaned and structured historical transactional dataset prepared as Apache Spark tables for scalable association analysis and saved within the company's Databricks framework.
- **Snowflake Database:** A comprehensive set of primary-to-secondary product mappings generated from the analysis and stored in a database for downstream use. Additionally, a list of primary-secondary product pairs that are not stocked in each inventory location was provided.

1.4 Results and Recommendations

The results of this thesis demonstrate that incorporating product relationships into inventory decision-making can significantly improve stocking effectiveness. By identifying complementary products and ensuring their joint availability, it is possible to better align inventory with observed demand patterns and reduce inefficiencies caused by incomplete product groupings.

Based on these findings, the following recommendations are proposed:

- Prioritize stocking of primary products alongside their associated secondary complements.
- Monitor products whose demand is highly dependent on specific co-purchased items.
- Identify and address gaps where frequently associated products are not jointly stocked.
- Incorporate association-based classifications into ongoing inventory planning processes.
- Use transaction data continuously to update and refine product relationship mappings.

1.5 Document Outline

The remainder of this thesis is organized as follows. Chapter 2 reviews relevant literature on the combination of association rule learning and clustering used in inventory optimization. Chapter 3 describes the data sources, preprocessing steps, and methodology used to construct the transactional dataset and create product mappings. Chapter 4 presents the results of the analysis, including extracted product relationships and classification outcomes. It also applies these results to inventory data to identify products at risk of obsolescence and evaluates the effectiveness of the proposed framework. Finally, Chapter 5 concludes the thesis with a summary of findings and directions for future work. Certain results in this document are proprietary to the partner company and have been redacted accordingly.

Chapter 2

Literature Review

This chapter reviews literature relevant to identifying relationships between products in retail transaction data and applying these relationships to support inventory decision-making. The review first discusses demand interdependence in retail inventory systems, followed by market basket analysis and association rule learning as methods for identifying co-purchasing relationships between products. Clustering methods for discovering higher-level product group structures are then examined. The chapter concludes by summarizing the limitations of existing approaches. This retail framework can then be expanded for wholesale applications. Together, this literature provides the foundation for developing a framework that identifies primary demand-generating products and their complementary secondary counterparts using transaction-level retail data.

2.1 Demand Interdependence in Retail Inventory Systems

Retail inventory decisions have traditionally been supported by product-level forecasting and classification methods that treat stock-keeping units (SKUs) as independent demand entities. However, existing literature demonstrates that consumer purchasing behaviour is frequently structured around groups of related products rather than isolated items. As a result, inventory performance depends not only on the availability of individual products but also on the joint availability of complementary items.

Research in retail inventory planning has shown that ignoring demand interdependencies among products can reduce retailers' profitability and lead to suboptimal stocking decisions. For example, Cachon and Kök (2007) show that inventory planning must account for basket-based purchasing behaviour because customers typically purchase a combination of products rather than individual

items independently. Hübner (2011) demonstrates that inventory planning, shelf-space allocation, and replenishment decisions are inherently linked across products, requiring models that capture these relationships. These studies highlight the importance of modelling demand relationships when designing inventory strategies.

2.2 Market Basket Analysis and Association Rule Learning

Market basket analysis (MBA) is a data mining approach for identifying demand relationships between products in retail transaction data. Rather than analyzing items independently, MBA examines combinations of products purchased together within individual transactions to uncover recurring patterns of co-occurrence (Kaur and Kang, 2016). Association rule learning (ARL) forms the analytical foundation of market basket analysis by identifying statistically significant relationships between products within transactional datasets (Agrawal and Srikant, 1994).

Studies have shown that association rules can reveal previously hidden purchasing regularities and support the identification of complementary product groupings within large assortments. For example, Brijs (2002) shows that association rule mining enables retailers to extract product interdependencies directly from transaction-level data and apply these relationships to category management and assortment optimization decisions. Similarly, Ali and Hammami (2022) demonstrate that ARL techniques such as Apriori and FP-Growth provide methods for identifying recurring product combinations. Together, these studies establish ARL as a reliable method for extracting actionable demand relationships from retail transaction data.

Other research has extended the use of ARL beyond descriptive pattern discovery to support retail decision-making. Venkatachari and Chandrasekaran (2016) show that association rules derived from historical transaction data can be used to identify relationships among products and improve assortment stocking strategies. These findings suggest that ARL provides a scalable framework for identifying structural relationships within large product portfolios.

2.3 Clustering Methods for Product Relationship Discovery

While association rule learning provides an effective method for identifying pairwise relationships between products, clustering techniques support the discovery of higher-level structure within retail assortments by grouping related products based on similarities in purchasing behaviour. Clustering has been widely applied to transactional retail datasets to reveal latent relationships between prod-

ucts and customer purchasing behaviour. Early work by Ghosh and Strehl (2005) demonstrates that clustering techniques applied to market basket data can identify meaningful product groupings in large transactional datasets.

Clustering methods can be integrated with ARL to improve the interpretation of the relationships generated by it. Liu et al. (2018) demonstrate that combining k-means clustering with ARL enables retailers to better understand purchasing behaviour and identify relationships. Similarly, Kanavos et al. (2025) show that hybrid frameworks combining clustering and ARL improve the extraction of behavioural patterns from transaction data and support downstream retail decision-making processes.

2.4 Limitations of Existing Approaches

Although MBA provides effective tools for identifying product relationships within transactional datasets, several limitations restrict their direct applicability to inventory decision-making. In particular, most ARL approaches focus on identifying statistically significant co-occurrence relationships between products without explicitly distinguishing between products that generate independent demand and those whose demand depends on complementary items. As a result, association rules often reveal relationships between products but do not provide classifications like we are seeking. Brijs et al. (1999) note that traditional assortment optimization approaches require estimates of substitution and complementarity effects between products, which are often difficult and costly to obtain in practice, limiting their applicability in real retail environments. They further argue that association rules become useful for assortment decision-making only when embedded within a microeconomic decision framework rather than used as a pattern-discovery tool.

Clustering-based approaches similarly provide useful insight into product similarity but often focus on identifying substitutable product groups rather than directional complementary relationships between items. For example, clustering techniques applied to retail transaction data are frequently used to support customer segmentation or inventory analysis (Ghosh and Strehl, 2005), but these approaches usually identify groups of related products without distinguishing which items act as primary demand drivers within those groups. Thus, clustering methods alone are not sufficient to support inventory decisions based on product relationships.

Taken together, these limitations indicate that although association rule learning and clustering techniques provide valuable insight into product relationships within retail assortments, existing approaches do not directly support the identification of demand-driving primary products and demand-

dependent complementary products for inventory planning purposes. The framework proposed in this study attempts to address this limitation by integrating ARL and clustering techniques to identify primary–secondary product relationships from transactional data.

2.5 Summary

The reviewed studies provide strong support for using a combination of association rule learning and clustering to uncover hidden product patterns. ARL helps identify which products are often purchased together, while clustering helps reveal natural groupings of items. However, several key gaps remain:

- ARL identifies connections but not product roles,
- clustering groups items by similarity rather than dependency,
- few studies combine both approaches to build hierarchical classifications in wholesale settings.

This study aims to close those gaps by developing a scalable, data-driven framework that integrates ARL and clustering to classify products into primary and secondary categories. The framework will adapt methods used in retail analytics to industrial wholesale environments and provide insights that improve stocking decisions and reduce product obsolescence.

Chapter 3

Methods

This chapter describes the methodology used to identify relationships between products and classify them according to their functional roles within the relationship. Section 3.1 introduces the datasets used in the analysis and outlines the preprocessing steps applied to prepare transaction-level data for modelling. Section 3.2 presents the modelling framework, beginning with the extraction of product relationships using association rule learning. It also highlights the application of clustering techniques to identify the product role. Finally, Section 3.3 describes how the outputs of these modelling approaches are integrated to produce primary-to-secondary mappings used for downstream inventory analysis.

3.1 Dataset

The dataset used in this study is proprietary and is sourced from the company's internal data repository. Among the available data tables, three were identified as essential for addressing the problem. These tables are:

- **Transaction Data:** The core dataset that contains over thirteen years of customer purchase records, including order identifiers, product identifiers, quantities, monetary values, and transaction timestamps.
- **Product Groupings:** A dataset that provides a higher-level categorization of products.
- **Inventory:** Product identifiers, descriptions, stock levels, and location information.

These three tables were combined as follows. The transaction data and product grouping tables were merged using a left merge on their shared item ID, which added the product classification to each

transaction. This was then left merged on the item ID with the inventory data to obtain the product descriptions. Following the merge, **Table 3.1** summarizes the key fields and their descriptions.

Table 3.1: Overview of data fields

Data Field	**Data Format**	**Description**
Order Number	Text	Unique order ID
Order Date	Date	Date on which order was made
Item Quantity	Number	The number of units bought
Sales Amount	Number	Total amount paid for purchased quantity
Product ID	Text	Unique product identifier
Product Group	Text	Item category the product falls into
Item Description	Text	Short description of the product

To ensure consistency in transaction-level analysis, a unique transaction identifier was constructed by concatenating the original order number, invoiced date, and customer identifier. This transformation ensures that each transaction reflects a distinct customer purchase event. The dataset was then filtered to include only transactions in the 2025 calendar year, providing a relevant and consistent time horizon for analysis.

Given the scale and diversity of the data, the dataset was partitioned by the product group into separate subsets. They are Waterworks, HVACR, Hydronics, and Plumbing. This segmentation allows for more targeted analysis within each product domain. When filtering by product group, any transaction containing a product from the target group (e.g., Waterworks) was retained in full, including all associated items, preserving cross-group relationships while allowing primary products within the group to be analyzed independently. Each subset was stored as an Apache Spark table to enable scalable processing and efficient downstream analysis.

After data cleaning, including the removal of null values and correction of data entry errors, the final dataset is comprised of 1.69×10^5 unique products across 9.21×10^6 item-level purchases.

The rest of this thesis focuses on the Waterworks subgroup, which provides a representative and sufficiently large sample for evaluating the proposed methodology. The Waterworks subset is comprised of 2.24×10^4 unique products across 4.50×10^5 item-level purchases.

3.2 Modelling

This section describes the modelling framework used to identify product relationships and classify products based on their role within transaction patterns. The approach consists of two main components: (i) product relationship modelling using ARL, and (ii) product clustering using derived network-based and product-level features.

3.2.1 Product Relationship Modelling

Product relationships were modelled using association rule learning (ARL), an unsupervised data mining technique designed to identify co-occurrence patterns within transactional datasets. Given a transaction database $\mathcal{D} = \{t_1, t_2, \ldots, t_n\}$, where $\mathcal{I}$ denotes the set of all items and each transaction $t_i \subseteq \mathcal{I}$ represents the set of items purchased in transaction i, an association rule is defined as an implication of the form $X \Rightarrow Y$, where $X, Y \subseteq \mathcal{I}$ and $X \cap Y = \emptyset$ (Agrawal and Srikant, 1994). Informally, this means that transactions containing all items in X tend to also contain all the items in Y, and X and Y do not contain common items. For this study, we will only look at $|X| = |Y| = 1$, meaning that both the antecedent X and the consequent Y consist of a single item. Here, $|\cdot|$ denotes the cardinality, i.e., the number of elements contained in the set. This restriction allows us to focus on simple pairwise associations that are easy to interpret and act upon. Additionally, it reduces computational complexity and avoids sparsity issues from larger, less frequent itemsets. The objective of ARL is to identify rules that satisfy user-defined thresholds on support and confidence (Agrawal et al., 1993).

Formally, the support of an itemset X is defined as

$$\text{support}(X) = \frac{|t \in \mathcal{D} : X \subseteq t|}{|\mathcal{D}|},$$

where $|\mathcal{D}|$ denotes the total number of transactions in the database and $|t \in \mathcal{D} : X \subseteq t|$ denotes the number of transactions containing X. Informally, support is the proportion of transactions that contain X. Similarly, the support of the joint itemset $X \cup Y$ is defined as

$$\text{support}(X \cup Y) = \frac{|t \in \mathcal{D} : X \cup Y \subseteq t|}{|\mathcal{D}|},$$

The support of the association rule $X \Rightarrow Y$ is given by

$$\text{support}(X \Rightarrow Y) = \text{support}(X \cup Y),$$

as the direction of the relationship does not affect the support. The support of a rule $X \Rightarrow Y$ is the proportion of transactions that contain both X and Y.

The confidence of a rule is defined as

$$\text{confidence}(X \Rightarrow Y) = \frac{\text{support}(X \cup Y)}{\text{support}(X)},$$

the probability that a transaction containing X also contains Y. Using confidence is standard in association analysis and provides us with an understanding of rule strength, statistical dependence, and predictive reliability (Han et al., 2011).

To further evaluate the strength and usefulness of associations, conviction was used. The conviction metric is defined by

$$\text{conviction}(X \Rightarrow Y) = \frac{1 - \text{support}(Y)}{1 - \text{confidence}(X \Rightarrow Y)},$$

how strongly the presence of X implies the presence of Y, taking into account how often Y appears overall. Conviction can range from 0 to infinity, where infinity means that the rule $X \Rightarrow Y$ is always true in the data. A conviction threshold of 1 was applied following standard practices, as values near 1 indicate independence and values larger than 1 indicate interesting rules (Azevedo and Jorge, 2007).

To illustrate these measures, consider a simple example transaction database

$$\mathcal{D} = \{t_1, t_2, t_3, t_4, t_5\},$$

where the item universe is

$$\mathcal{I} = \{\text{bread, milk, butter}\},$$

and the transactions are:

$$t_1 = \{\text{bread, milk}\}$$
$$t_2 = \{\text{bread, butter}\}$$
$$t_3 = \{\text{milk, butter}\}$$
$$t_4 = \{\text{bread, milk}\}$$
$$t_5 = \{\text{bread, milk, butter}\}.$$

Consider the association rule

$$\{\text{bread}\} \Rightarrow \{\text{milk}\}.$$

The support of the joint itemset is

$$\text{support}(\{\text{bread}\} \cup \{\text{milk}\}) = \frac{3}{5} = 0.6,$$

since three transactions contain both items out of the total five. The support of the antecedent, {bread}, is

$$\text{support}(\{\text{bread}\}) = \frac{4}{5} = 0.8.$$

The confidence of the rule is

$$\text{confidence}(\{\text{bread}\} \Rightarrow \{\text{milk}\}) = \frac{0.6}{0.8} = 0.75,$$

meaning that 75% of transactions containing bread also contain milk.

The conviction of the rule is

$$\text{conviction}(\{\text{bread}\} \Rightarrow \{\text{milk}\}) = \frac{1 - 0.8}{1 - 0.75} = 0.8.$$

Because the conviction value is less than 1, we can say that the association rule {bread} ⇒ {milk} is weak and does not suggest a meaningful relationship.

The modelling process follows a two-step decomposition: (i) identification of frequent itemsets satisfying a minimum support threshold, and (ii) generation of association rules from these itemsets using confidence constraints (Agrawal and Srikant, 1994). There are two common ARL algorithms, Apriori and FP-Growth. In this study, frequent itemsets were mined using the FP-Growth algorithm, which was chosen because of its computational efficiency with larger datasets. By leveraging a compressed tree-based representation of the data, FP-Growth outperforms Apriori (more detailed information can be found in (Han et al., 2011)).

Formally, the FP-Growth algorithm identifies frequent itemsets by constructing a compressed representation of the transaction database known as a frequent pattern tree (FP-tree). The FP-tree is a prefix-tree structure that stores items appearing in the dataset together with their frequencies in a hierarchical format, allowing shared prefixes between transactions to be represented only once. To construct the FP-tree, the database is first scanned to determine the frequency of each item and retain only those items whose support exceeds the minimum support threshold. The transactions are then reordered according to decreasing item frequency and inserted into the tree structure, where overlapping item sequences share common branches.

From the previous example, using the transaction database defined above, item supports are first computed, and the items are ordered in descending frequency as

$$\text{bread} \rightarrow \text{milk} \rightarrow \text{butter}.$$

Bread was chosen as the first node because it is tied with milk as the most frequent item (each appearing 4 times), and the tie was resolved using a lexicographical ordering rule. Each transaction is then reordered according to this frequency and inserted into the FP-tree. The resulting FP-tree

is shown in **Figure 3.1**. Here, each node stores an item label together with its frequency count, representing how many transactions share that prefix path.

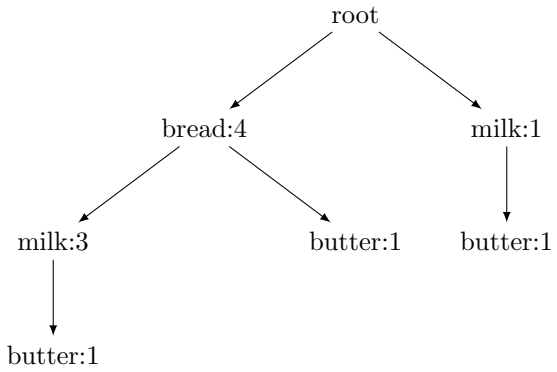


Figure 3.1: FP-tree example structure

After the FP-tree has been constructed, frequent itemsets are extracted recursively by exploring conditional subsets of the tree associated with each item. These conditional subsets represent transactions that contain a given item and allow the algorithm to identify larger frequent item combinations efficiently without repeatedly scanning the original transaction database. By storing the dataset in this compressed tree structure and mining frequent itemsets directly from it, the FP-Growth algorithm reduces both the number of database scans and the computational effort required to identify frequent itemsets, making it well-suited for analyzing large transactional datasets (Han et al., 2011). FP-Growth was conducted in Python with products represented as columns and transactions represented as rows using the `mlxtend` library (Raschka, 2018). Before model execution, transactions were filtered, removing infrequent items (occurring fewer than five times) to reduce noise and computational complexity. A one-hot encoded matrix was created with products represented as columns and transactions as rows using the `TransactionEncoder` implementation from `mlxtend`. In this representation, each cell takes the value `True` if a product is present in the transaction corresponding to the row and `False` otherwise. For example, consider a shortened version of the above example where

$$\mathcal{D} = \{\{\text{Milk, Bread}\}, \{\text{Bread, Butter}\}, \{\text{Milk, Butter}\}\}.$$

	Bread	Milk	Buter
t_1	True	True	False
t_2	True	False	True
t_3	False	True	True

Table 3.2: Example of a one-hot encoded transaction matrix

After applying one-hot encoding, the dataset is transformed into a binary matrix, which can be seen in **Table 3.2**. This table provides the binary transaction representation required for frequent itemset mining using the FP-Growth algorithm.

Association rules were generated using the FP-Growth algorithm, and conviction was used as the primary metric. The rules were subsequently filtered using a confidence threshold to be established based on the results. These thresholds were selected to balance coverage and reliability of the resulting relationships. The final output of this stage is a directed set of product relationships, where antecedents represent candidate primary products, and consequents represent their associated secondary products. ARL has been widely applied in retail and recommendation systems to uncover item-to-item relationships and support decision-making in merchandising and inventory management (Kaminskas et al., 2017). In this context, the extracted rules provide the foundational structure for identifying complementary product relationships, which are further refined in subsequent modelling steps.

3.2.2 Product Clustering

In addition to pairwise product relationships, clustering was employed to identify groups of products with similar structural characteristics. This allows us to identify which products within these itemsets are the primary items of a sale and which ones are their related secondary items. Clustering is an unsupervised machine learning technique that partitions a dataset into homogeneous groups based on feature similarity (Mehlstäubl et al., 2023).

Each product was represented as a 10-dimensional feature vector derived from both transactional characteristics and network-based structural attributes, which served as inputs to the clustering algorithms. There are three network-based features and seven transactional features. A breakdown of these features into their respective network-based and transactional components is provided below.

To derive the network features, a directed product network was constructed from association rules, in which nodes represent products and edges represent directional relationships from antecedent to consequent. The node set V consisted of all products appearing in the filtered association-rule dataset, while the edge set E was defined by ordered pairs (i, j) corresponding to rules of the form $i \Rightarrow j$. A directed graph $G = (V, E)$ was then created using the `NetworkX` Python library (Hagberg et al., 2008). From this directed graph, several centrality measures were computed for each product. They are:

- **Degree:** the number of direct connections a product has to other products in the association-

rule network. For this metric, the association-rule network is treated as an undirected graph, so that a connection between two products is counted regardless of the direction of the rule ($X \Rightarrow Y$ or $Y \Rightarrow X$). The degree of node i in a graph with adjacency matrix A is

$$k_i = \sum_{j=1}^{n} A_{ij},$$

where $A_{ij} = 1$ if nodes i and j are connected, and 0 otherwise. The diagonal elements of the adjacency matrix are set to zero ($A_{ii} = 0$), since self-loops are not permitted in the association-rule network because rules are defined only between distinct products.

- **Closeness Centrality:** how close a product is, on average, to all other products in the network based on shortest-path distance. For a directed graph, the closeness centrality of node i is defined as

$$C_C(i) = \frac{n-1}{\sum_{j \neq i} d(i,j)},$$

where $d(i,j)$ denotes the length of the shortest directed path from node i to node j, and n is the total number of nodes in the network. Because the association-rule network is directed, shortest paths follow the direction of rules ($X \Rightarrow Y$).

- **Eigenvector Centrality:** measures the importance of a product within the network by assigning higher scores to products that are connected to other highly connected products. Formally, the eigenvector centrality of node i is defined as

$$x_i = \frac{1}{\lambda} \sum_{j=1}^{n} A_{ij} x_j,$$

where A is the adjacency matrix of the network, x_i is the eigenvector centrality score of node i, and λ is the largest eigenvalue of A. Nodes with higher values of x_i are therefore those connected to other nodes that are themselves highly central within the association-rule network.

Because the association-rule network may consist of multiple disconnected components after applying support, confidence, and conviction thresholds, shortest paths do not necessarily exist between all pairs of nodes. Therefore, closeness centrality was computed using only nodes reachable through directed paths from each product, following the implementation provided in `NetworkX`. Similarly, eigenvector centrality reflects the relative importance of products within their connected components rather than across the entire network. These metrics capture the structural importance of products

within the network, reflecting their role in co-purchase relationships (Rout et al., 2024). Additionally, various product-level features were computed. A full list can be found in **Table 3.3**. These engineered features represent local, product-level transactional behaviour.

Table 3.3: Overview of product level features

Feature	Description
Derived Price	The price per unit computed by dividing the total transaction amount by the number of units purchased.
Co-purchase Diversity	The number of distinct products co-purchased with a given product across all transactions.
Normalized Co-purchase Diversity	The ratio of distinct co-purchased products to distinct product groups.
Average Basket Size	The average number of products appearing in transactions that contain the given product.
Product Frequency	The number of transactions containing the product.
Dominant Product Group Ratio	The proportion of transactions in which the product is most frequently associated with its dominant product group.
Average Purchased	The average quantity purchased per transaction for each product (capped at 200 units to limit the influence of extreme values).

Two clustering algorithms were considered: K-Means (Lloyd, 1957; MacQueen, 1967) and Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) (McInnes et al., 2017), each selected based on its complementary strengths. The K-Means algorithm partitions data into K clusters by minimizing within-cluster variance, formally defined as:

$$min \sum_{i=1}^{K} \sum_{x \in C_i} ||x - \mu_i||^2,$$

where x represents the feature vector describing a product in the constructed feature space, C_i denotes the set of observations assigned to cluster i, and μ_i represents the centroid (mean vector) of cluster C_i. K-means finds K cluster centres that minimize the sum of squared distances between each point and its nearest centre. K-Means is widely used due to its computational efficiency, scalability, and ability to handle large datasets, making it suitable for high-volume transactional data (Abbasian and Jamili, 2025). K-means requires the number of clusters to be set before being run. We can address this limitation by employing the elbow method (Thorndike, 1953). It plots the within-cluster sum of squares as a function of K and identifies the point at which additional clusters produce diminishing reductions in variance. This point represents a balance between model simplicity and clustering accuracy, helping to select a value of K that captures meaningful structure

in the data without over-partitioning the feature space.

K-means assumes that clusters are approximately spherical and of similar size, which may limit its effectiveness when applied to datasets with irregular cluster structures. To address these limitations, the HDBSCAN algorithm was also employed. As a density-based clustering method, HDBSCAN is capable of identifying clusters with varying shapes and densities while explicitly detecting noise points, making it particularly well-suited for complex product distributions (McInnes et al., 2017).

For each product subgroup, both algorithms were evaluated, and the clustering method was selected based on the quality and interpretability of the resulting clusters. This follows established practices of clustering complex data, where multiple algorithms are compared to determine the most appropriate representation of the data structure (Mehlstäubl et al., 2023).

Due to the lack of labelled data and the unsupervised nature of clustering, clusters were interpreted qualitatively. Clusters were evaluated based on their alignment with the directed product network derived from association rules. They were further validated through consultation with a product expert, ensuring that the identified groupings were consistent with domain knowledge and real-world product relationships. We were provided with a small set of identified primary products, which were then used to determine the cluster that represented the primary group.

The final clustering output organizes products into behaviourally similar groups, complementing the pairwise associations identified through ARL and supporting the classification of products into functional roles within the inventory. With this, a single cluster was isolated as the *primary* group, and the products within this group were treated as *primary* products.

3.3 Model Integration and Product Classification

The final stage of the modelling framework integrates the outputs of ARL and product clustering to classify products into functional roles within the inventory. This integration combines pairwise co-purchase relationships with higher-level structural groupings to produce a more robust and interpretable classification. Products are first identified as primary or secondary based on the clustering results, and association rules are then used to establish directional dependencies between products. This is illustrated in **Figure 3.2**. Specifically, the transaction data is used to create the association-rule network, and the results from the network are combined with the data to perform clustering. The outputs of both are then combined to create the final mapping of primary to secondary products.

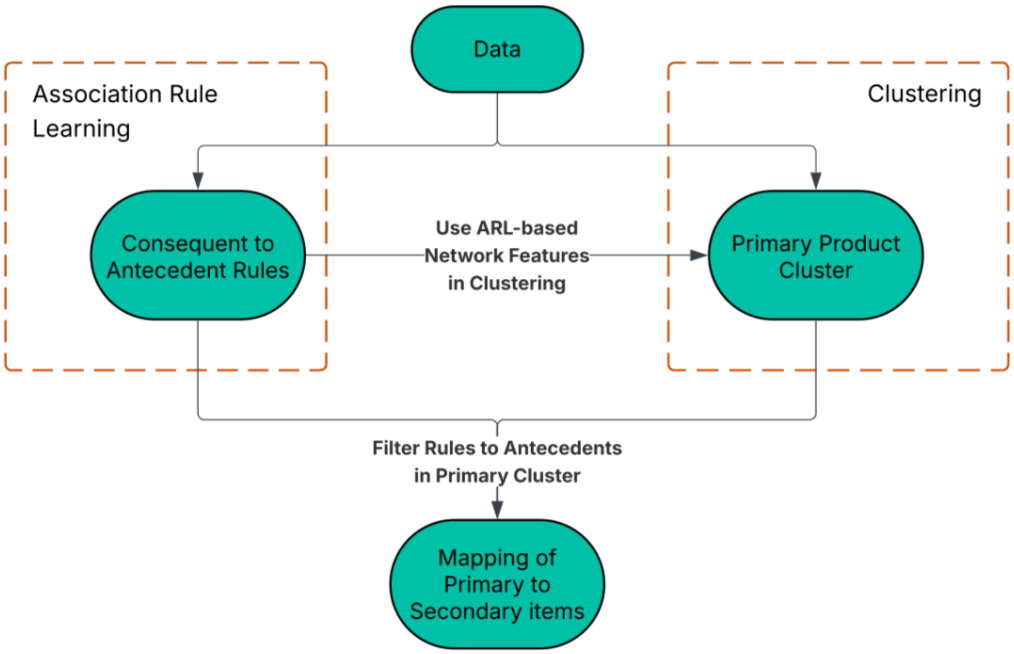


Figure 3.2: Product mapping methodology

Products belonging to the cluster identified as the *primary* group were treated as anchor nodes within the association-rule network. Let V be the set of nodes in the association-rule network, let $P \subseteq V$ denote the set of products assigned to the primary cluster, and let $E = \{(i, j)\}$ denote the set of directed association rules obtained from ARL. The final mapping was constructed by assigning a directional dependency between products whenever an edge $(i, j) \in E$ existed such that $i \in P$. Formally, a dependency relationship between products i and j was retained in the final classification if

$$(i, j) \in E \quad \text{and} \quad i \in P.$$

This combined framework ensures that classification is not dependent on local co-purchase relationships or global clustering structure alone, but instead leverages both to produce a more accurate representation of product roles. The final product is a mapping of primary products to their corresponding secondary products, ranked by the confidence scores of the association rules. Results will be presented in Chapter 4.

3.4 Summary

This chapter presents the methodological framework for identifying product relationships and classifying items by their roles in observed purchasing patterns. Because the modelling approach relies primarily on unsupervised learning techniques, formal quantitative evaluation of classification accuracy is inherently limited; therefore, model performance was assessed by the consistency of the association rules, the interpretability of the clustering results, and validation through consultation with product experts at the company to ensure alignment with real-world product dependencies. To summarize, the work done includes preparing a transaction-level dataset for analysis, extracting directed product relationships using FP-Growth, clustering products with HDBSCAN and K-means using engineered transactional and network-based features, and integrating these results to produce primary–secondary product mappings used for downstream inventory analysis.

The rest of this thesis will primarily include results from the waterworks subgroup. This subset of data provides a representative and sufficiently large sample for evaluating the proposed methodology while maintaining consistency in product structure and purchasing patterns across the analysis. The methodology has been extended to the other product groups; however, for the sake of this study, the analysis is limited to the waterworks subgroup to provide a clear evaluation of the approach and maintain consistency in the presentation of results.

Chapter 4

Results

This chapter presents the results of the combined association rule learning and clustering framework developed in Chapter 3. The analysis proceeds in four stages: (i) the results of the association rule learning approach are examined to identify product relationship patterns extracted from transactional data, (ii) the clustering results are analysed to assess the structure of product groupings, (iii) the final classification of products into primary and secondary roles is presented and evaluated, and (iv) the primary-to-secondary product mappings are applied to current inventory data to identify locations where complementary products are missing alongside stocked primary items.

4.1 Evaluation of Modelling Approaches

4.1.1 Association Rule Learning Results

Association Rule Learning (ARL) was applied to identify directional co-purchase relationships between products. Using the FP-Growth algorithm with a confidence threshold greater than 0.33 and additional filtering based on conviction, a large set of pairwise relationships was extracted, describing how products are jointly purchased within transactions. Note: the 0.33 threshold was set specifically for the waterworks subset and will most likely be different for other product groups. This threshold for the Waterworks group was selected based on recommendations from a product expert at the company, who reviewed the generated product mappings and identified which associations were meaningful in practice; the final value was then determined based on the set of rules they considered acceptable.

The directional rules were used to construct a directed product relationship graph using the Net-

workX library, where nodes represent products and edges represent association rules. Centrality and connectedness measures computed from this graph were used in downstream clustering. The network structure created from the association rules is discussed in Section 4.1.2.

Table 4.1: Examples of association rules with metrics (product codes have been hidden)

Antecedents	**Consequents**	**Antecedent Support**	**Consequent Support**	**Support**	**Confidence**	**Lift**	**Conviction**
A	B	0.000147	0.00129	0.00011	0.750	579.606	3.995
A	C	0.000147	0.00649	0.00011	0.750	115.593	3.974
A	D	0.000147	0.00273	0.000092	0.625	229.306	2.659
B	A	0.00129	0.000147	0.00011	0.0851	579.606	1.0929
C	A	0.00649	0.000147	0.00011	0.0170	115.593	1.0171

The resulting association rules for waterworks products consisted of 1.514×10^6 rules. **Table 4.1** provides several examples of the derived association rules, including their computed metrics. A representative example is the rule

$$\texttt{A} \Rightarrow \texttt{B},$$

where `A` is a hydrant and `B` is a mechanical joint gasket. This rule has a confidence value of 0.750, indicating that 75% of transactions containing the hydrant also include the gasket. The conviction value of 3.995, which is greater than 1, supports a strong directional dependency between the items. In contrast, the reverse rule

$$\texttt{B} \Rightarrow \texttt{A}$$

has a substantially lower confidence value of 0.0851 while still surpassing the conviction threshold. The low confidence score indicates that although gaskets are frequently purchased alongside hydrants, they are also commonly purchased in other contexts, and $\texttt{B} \Rightarrow \texttt{A}$ should not be considered an association rule. This observation supports the decision to set a threshold for confidence.

This forms the foundation of the product relationship modelling framework and provides interpretable mappings between frequently co-purchased items. When combined with the clustering results, these mappings can be further refined to retain meaningful primary-to-secondary relationships, improving the overall clarity and usefulness of the product association structure.

4.1.2 Clustering Results

Clustering was applied to engineered product-level features derived from transactional data and network structure. These features included the derived price per unit, average quantity purchased, purchase frequency, diversity in co-purchased products, average basket size in transactions, and

dominant product group concentration, as described in Chapter 3.

The association-rule network constructed for the waterworks category consisted of 8,562 product nodes and 1,514,780 directed edges, as summarized in **Table 4.2**. Here, nodes represent individual products appearing in the rule set, and directed edges represent association rules linking antecedent products to consequents. The network formed three weakly connected components, where weak connectivity refers to groups of products that remain connected when rule direction is ignored, with the largest component containing 8,554 products, indicating that nearly all items participate in a shared co-purchase structure. The average degree of 353.84 indicates that each product was connected to approximately 354 other products through association rules on average, while the mean in-degree and out-degree values of 176.92 represent the average number of incoming and outgoing rule relationships per product, respectively. Despite the large number of edges, the overall network density was 0.0207, meaning that only a small proportion of all possible product-to-product relationships were present, indicating a sparse but highly interconnected association-rule structure.

Table 4.2: Structural summary of the waterworks association-rule network

Metric	**Value**
Number of nodes	8,562
Number of edges	1,514,780
Network density	0.0207
Average degree	353.84
Average in-degree	176.92
Average out-degree	176.92
Weakly connected components	3
Strongly connected components	3
Largest component size	8,554

Both K-Means and HDBSCAN clustering algorithms were evaluated for each product group. The final clustering method selected for each product group was the one that produced the most interpretable separation between the two classes of products compared to the structure of the association-rule network and based on the given primary products.

For the waterworks category, the engineered and given features were projected into a two-dimensional representation using t-distributed stochastic neighbour embedding (TSNE) from the `scikit-learn` Python library (Maaten and Hinton, 2008), as shown in **Figure 4.1**. A set of products independently identified as primary based on domain knowledge was overlaid on this visualization to provide a reference for evaluating clustering performance. These known primary products were observed to concentrate within a distinct region of the feature space, primarily located in the lower-left portion of the TSNE projection, suggesting that primary products share similar structural characteristics in

the engineered feature space.

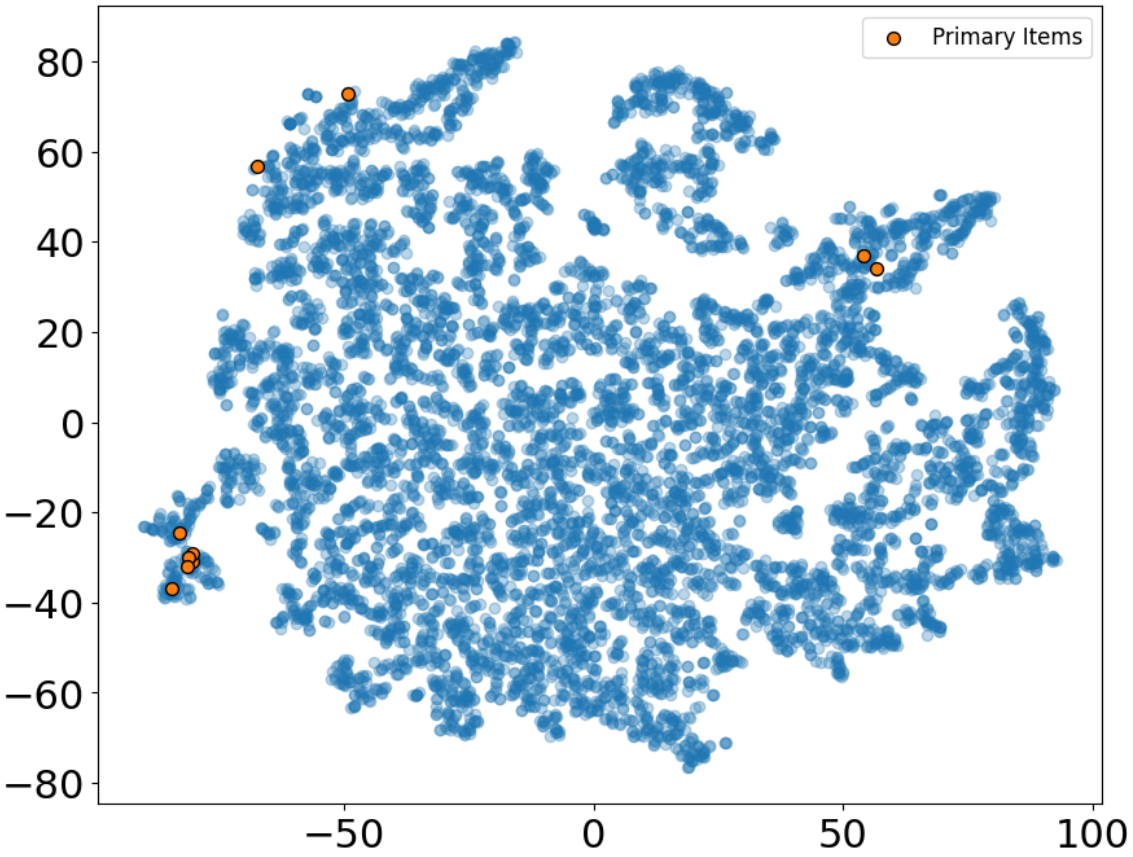


Figure 4.1: TSNE visualization of products

To determine which cluster corresponded most closely to the primary product role, clustering results from both HDBSCAN and K-means were compared against this reference distribution. The number of clusters for the K-means algorithm was selected using the elbow method, which indicated an optimal value of approximately $K = 9$ based on diminishing reductions in within-cluster sum of squares beyond this point. As shown in **Figure 4.2**, Cluster 1 identified by HDBSCAN aligns closely with the region containing the majority of the known primary products. Similarly, in the K-means results shown in **Figure 4.3**, Cluster 6 captures a comparable grouping.

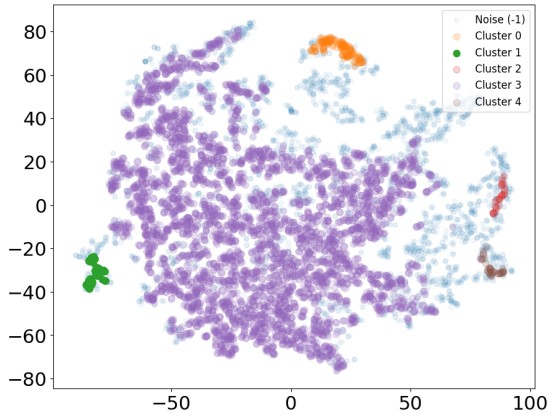


Figure 4.2: HDBSCAN clustering of waterworks products

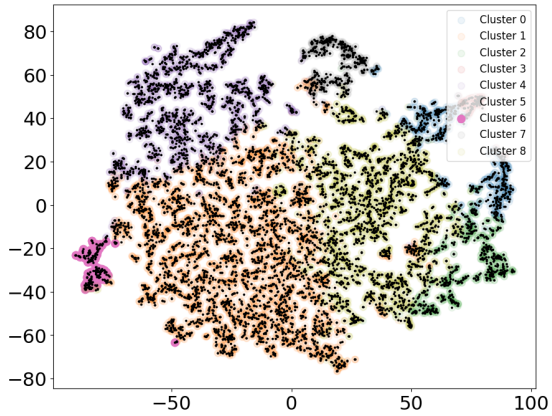


Figure 4.3: K-means clustering of waterworks products

The HDBSCAN cluster was chosen over the K-means cluster for the waterworks subset because HDBSCAN has the added benefit of filtering out noise compared to K-means. If the clusters are roughly the same, HDBSCAN is more likely to have a group of products that belong to the true primary cluster rather than K-means because of its ability to remove noise (Ibraimoh and Aderoba, 2025). Therefore, Cluster 1 from the HDBSCAN clustering was chosen as the primary product cluster. Across product groups, clustering consistently revealed two distinct structural product roles: lower-centrality products with a higher price, and higher-centrality products with broader co-purchase connectivity and a lower price.

4.1.3 Final Product Classification Results

The outputs of association rule learning and clustering were integrated to classify products into primary and secondary roles. Products that consistently appear as antecedents in high-confidence rules and belong to clusters with higher unit prices, lower average purchase quantities, and lower degree values, indicating fewer connections to other products in the network structure, tend to be primary products. Conversely, products that more frequently appear as consequents in association rules and belong to clusters characterized by lower unit prices, higher average purchase quantities, and higher degree values—indicating connections to a larger number of other products in the network structure—tend to be secondary products. 53 primary products and 4982 secondary products were identified. As shown in **Table 4.3**, products in the primary cluster exhibit substantially higher average prices and lower purchase quantities than non-primary products, while also having fewer network connections on average, supporting their interpretation as anchor items around which complementary products are purchased.

Table 4.3: Overview of HDBSCAN clustering results of waterworks products

	Primary	**Non-Primary**
Count	53	4982
Derived Price	$4387	$170
Degree	176.302	207.943
Average Quantity per Transaction	1.686	9.420

One example of an identified primary product from the waterworks inventory is the item [redacted], which is a fire hydrant. This product has a derived price of $3474, a degree value of 120, and was purchased as a single item on average, all of which are consistent with the derived characteristics of the primary cluster.

The result of combining the association-rules network and clustering results is a mapping from primary products to their related secondary products. Each mapping is paired with a confidence score so that additional filtration may be done if needed. Eleven secondary products were identified as being associated with the previously classified primary product [redacted]. These products have an average derived price of $145, a substantially higher average degree of 1716, and were purchased in quantities of approximately 26 units per transaction on average. The associated items include joint fittings for hydrants, pipe preparation tools, and hydrant tees, all of which function as complementary components required during hydrant installation and maintenance. Their lower prices, higher connectivity within the association-rule network, and larger purchase quantities are consistent with the characteristics of secondary products shown in **Table 4.3**, supporting the interpretation that these items depend on the selection of the primary product and therefore occupy supporting roles within the inventory structure.

Again, due to the unsupervised nature of association rule learning and clustering, as well as the absence of pre-labelled data, model evaluation could only be performed qualitatively. The resulting mapping was reviewed by a product expert from the company, who assessed whether the identified relationships and product role assignments were consistent with expected purchasing behaviour using their practical inventory knowledge. The expert confirmed that the majority of the identified primary–secondary product relationships aligned with typical installation workflows and stocking practices observed in the waterworks category, providing practical support for the validity of the modelling approach.

4.2 Inventory Alignment Using Product Mappings

The final stage of the analysis applied the primary-to-secondary product mappings to the company's current inventory data to identify locations where primary products are stocked without their associated complementary products. Inventory data was first filtered to retain the most recent inventory snapshot per product-location pair. The resulting dataset represents the current stocking configuration across inventory stores. For each location, the presence of primary products was compared with the availability of their corresponding secondary products. Locations where primary products were stocked, but one or more associated secondary products were not available, were flagged. The resulting output provides a structured list of product-location combinations in which stocking decisions may benefit from adjustment to better support observed purchasing patterns.

For the waterworks subset, ■ instances of primary products being stocked without their associated secondary products were identified across the analyzed locations. Identifying these mismatches provides actionable insight into locations where adjusting inventory composition may improve product availability for customers and prevent these primary products from becoming obsolete. The final list of product-location combinations was delivered to the company for future review and use in supporting inventory planning and stocking decisions.

4.3 Summary

This chapter presents the results of the combined ARL and clustering framework used to classify products into functional inventory roles and their relationships. ARL extracted over 1.5 million directional co-purchase relationships from transactional data, forming a product relationship network that captures complementary purchasing behaviour.

Clustering was then applied to product-level features derived from transactional behaviour and network structure. HDBSCAN was selected as the preferred clustering method for the waterworks subgroup due to its ability to separate meaningful structure while filtering noise. Across product groups, clustering consistently revealed two distinct product roles: higher-priced, low connectivity products that function as primary demand drivers, and lower-priced, more highly connected products that act as complementary secondary items.

The outputs of ARL and clustering were integrated to classify products into primary and secondary roles and to generate mappings between them. These mappings were qualitatively evaluated by a product expert from the company.

Finally, the primary-to-secondary product mappings were applied to current inventory data to identify locations where primary products were stocked without their associated complementary items. This produced a structured list of stocking gaps that can be used for future inventory decisions.

Chapter 5

Conclusion

This chapter outlines the key findings of the thesis, presents recommendations based on the modelling results, and identifies opportunities for extending the proposed framework in future work.

5.1 Discussion

This thesis developed a data-driven framework for identifying demand-driving products stocked in inventory without their related dependent secondary products. This will help the company flag items that are at risk of sitting on shelves and becoming obsolete. Functional relationships between products were found using historical transaction data and association rule learning. By creating a directed product relationship network and combining it with clustering based on transactional and network-derived features, products were classified into primary demand-driving items and secondary complementary items. The resulting mappings provide a structured representation of product dependencies that are not captured in traditional inventory classification systems.

The results demonstrate that ARL can be used effectively to uncover interpretable co-purchase relationships at scale, and that these relationships can be strengthened through the integration of graph-based centrality measures and clustering techniques. The combined modelling approach allowed us to identify structural differences between products that generate demand independently and those that depend on complementary availability.

More broadly, the framework demonstrates how unsupervised learning methods can be applied to large-scale transactional data to generate interpretable insights that support inventory decision-making. Rather than replacing existing stocking strategies, the approach provides an additional analytical layer that helps identify hidden dependencies between products and supports more in-

formed operational planning.

5.2 Recommendations

Based on the results of this study, several recommendations can be made for improving inventory alignment with observed purchasing behaviour.

- Primary–secondary product mappings should be incorporated into stocking decision workflows to ensure that complementary items are available alongside demand-driving products. This ensures the joint availability of frequently associated products and may help reduce missed sales opportunities caused by incomplete product groupings, as well as prevent items from sitting too long on shelves and becoming obsolete.
- The framework developed in this study should be integrated into existing analytics pipelines so that the product relationship mapping can be updated periodically as purchasing behaviour evolves and inventory changes. Maintaining this will allow stocking strategies to remain responsive to changes in customer demand.
- The primary–secondary product mappings should be used to support location-level inventory monitoring by identifying stores where complementary products are missing alongside stocked primary items. Regular review of these gaps can help prioritize targeted replenishment actions and improve the consistency of product availability across locations.

5.3 Next Steps

Several opportunities exist to extend the methodology developed in this thesis.

- The analysis could be expanded beyond pairwise relationships to incorporate higher-order itemsets, allowing more complex purchasing structures involving multiple complementary products to be captured. This would provide a more complete representation of transaction-level dependencies.
- The clustering framework could be extended through the inclusion of additional behavioural features or alternative graph-embedding techniques to further improve separation between product roles.

- The methodology could be integrated into a continuous monitoring workflow that automatically detects locations where primary products risk obsolescence, enabling proactive intervention before inventory issues develop further.

References

Abbasian, Mahyar and Amin Jamili (2025). "A hybrid machine learning approach to evaluate and select agile-resilient sustainable suppliers considering Supply Chain 4.0: A real case study". In: *Process Integration and Optimization for Sustainability*, pp. 717–735. DOI: 10.1007/s41660-025-00483-1.

Agrawal, Rakesh, Tomasz Imieliński, and Arun Swami (1993). "Mining association rules between sets of items in large databases". In: *ACM SIGMOD Record* 22, pp. 207–216. DOI: 10.1145/170036.170072.

Agrawal, Rakesh and Ramakrishnan Srikant (1994). "Fast Algorithms for Mining Association Rules in Large Databases". In: *Proceedings of the 20th International Conference on Very Large Data Bases (VLDB '94)*, pp. 487–499.

Ali, Mohamed Amine Ben and Oussama Hammami (2022). "Association Rule Mining for Market Basket Analysis in Retail Data: Enhancing Automated Knowledge Discovery with Apriori and FP-Growth Algorithms". In: *Journal of Data Mining, Knowledge Discovery, and Decision Support Systems* 12.3, pp. 1–18.

Azevedo, Paulo J. and Alípio M. Jorge (2007). "Comparing Rule Measures for Predictive Association Rules". In: *Machine Learning: ECML 2007*. Ed. by Joost N. Kok et al. Berlin, Heidelberg: Springer Berlin Heidelberg, pp. 510–517.

Brijs, Tom (2002). "Retail market basket analysis: a quantitative modelling approach". Phd thesis. Limburg University Center.

Brijs, Tom et al. (1999). "Using association rules for product assortment decisions: a case study". In: *KDD '99: Proceedings of the fifth ACM SIGKDD international conference on Knowledge discovery and data mining*, pp. 254–260. DOI: 10.1145/312129.312241.

Cachon, Gérard P. and A. Gürhan Kök (2007). "Category Management and Coordination in Retail Assortment Planning in the Presence of Basket Shopping Consumers". In: *Management Science* 53.6, pp. 934–951. ISSN: 00251909, 15265501.

Ghosh, Joydeep and Alexander Strehl (2005). "Clustering and Visualization of Retail Market Baskets". In: *Advanced Techniques in Knowledge Discovery and Data Mining*. Springer London, pp. 75–102. ISBN: 978-1-84628-183-9. DOI: 10.1007/1-84628-183-0_3.

Hagberg, Aric, Pieter J. Swart, and Daniel A. Schult (Jan. 2008). "Exploring network structure, dynamics, and function using NetworkX". In: Los Alamos National Laboratory (LANL). URL: https://www.osti.gov/biblio/960616.

Han, Jiawei, Micheline Kamber, and Jian Pei (2011). *Data Mining: Concepts and Techniques (3rd ed.)* Morgan Kaufmann.

Hübner, Alexander (2011). *Retail Category Management: Decision Support Systems for Assortment, Shelf Space, Inventory and Price Planning*. 1st ed. Lecture Notes in Economics and Mathematical Systems. Springer Berlin Heidelberg, pp. 1, 153. ISBN: 978-3-642-22477-5. DOI: 10.1007/978-3-642-22477-5.

Ibraimoh, Raphael and Adetunji Aderoba (Jan. 2025). "Comparison of K-Means and HDBSCAN Clustering Approaches to Enhance Marketing Strategies". In: *Iconic Research AND Engineering Journals*, pp. 272–282.

Kaminskas, Marius et al. (2017). "Product-Seeded and Basket-Seeded Recommendations for Small-Scale Retailers". In: *Journal on Data Semantics* 6, pp. 3–14. DOI: 10.1007/s13740-016-0058-3.

Kanavos, Andreas, Gerasimos Vonitsanos, and Phivos Mylonas (2025). "Integrating Machine Learning Approaches for Consumer Behavior Analysis in Retail Transactions". In: *2025 20th International Workshop on Semantic and Social Media Adaptation and Personalization (SMAP)*, pp. 7–12. DOI: 10.1109/SMAP66932.2025.00010.

Kaur, Manpreet and Shivani Kang (2016). "Market Basket Analysis: Identify the Changing Trends of Market Data Using Association Rule Mining". In: *Procedia Computer Science* 85. International Conference on Computational Modelling and Security (CMS 2016), pp. 78–85. ISSN: 1877-0509. DOI: https://doi.org/10.1016/j.procs.2016.05.180.

Liu, Run-Qing, Young-Chan Lee, and Hong Mu (2018). "Customer Classification and Market Basket Analysis Using K-Means Clustering and Association Rules: Evidence from Distribution Big Data of Korean Retailing Company". In: *Knowledge Management Research* 19, pp. 59–76. DOI: 10.15813/kmr.2018.19.4.004.

Lloyd, S. (1982). "Least squares quantization in PCM". In: *IEEE Transactions on Information Theory* 28.2, pp. 129–137. DOI: 10.1109/TIT.1982.1056489.

Maaten, Laurens van der and Geoffrey Hinton (2008). "Visualizing Data using t-SNE". In: *Journal of Machine Learning Research* 9.86, pp. 2579–2605. URL: http://jmlr.org/papers/v9/vandermaaten08a.html.

MacQueen, J. (1967). "Some Methods for Classification and Analysis of Multivariate Observations". In: *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability*. Ed. by L. M. Le Cam and J. Neyman. Vol. 1. University of California Press, pp. 281–297.

Martin, Travis, Xiao Zhang, and M. E. J. Newman (2014). "Localization and centrality in networks". In: *Physical Review E* 90.5. ISSN: 1550-2376. DOI: 10.1103/physreve.90.052808.

McInnes, Leland, John Healy, and Steve Astels (2017). "hdbscan: Hierarchical density based clustering". In: *Journal of Open Source Software* 2.11, p. 205. DOI: 10.21105/joss.00205.

Mehlstäubl, Jan et al. (2023). "Methodical Approach To Cluster Configurations Of Product Variants Of Complex Product Portfolios". In: *Proceedings of the Design Society* 3, pp. 2645–2654. DOI: 10.1017/pds.2023.265.

Raschka, Sebastian (2018). "MLxtend: Providing machine learning and data science utilities and extensions to Python's scientific computing stack". In: *Journal of Open Source Software* 3.24, p. 638. DOI: 10.21105/joss.00638.

Rout, Trilochan et al. (2024). "Centrality Measures and Their Applications in Network Analysis: Unveiling Important Elements and Their Impact". In: *Procedia Computer Science* 235. International Conference on Machine Learning and Data Engineering (ICMLDE 2023), pp. 2756–2765. ISSN: 1877-0509. DOI: https://doi.org/10.1016/j.procs.2024.04.260.

Thorndike, Robert L. (1953). "Who belongs in the family?" In: *Psychometrika* 18, pp. 267–276. URL: https://api.semanticscholar.org/CorpusID:120467216.

Venkatachari, Kavitha and Issac Davanbu Chandrasekaran (2016). "Market Basket Analysis Using FP-Growth and Apriori Algorithm: A Case Study of Mumbai Retail Store". In: *BVIMSR's Journal of Management Research* 8.1, pp. 56–63.